

Git is the free and open source distributed version control system that's responsible for everything GitHub related that happens locally on your computer. This cheat sheet features the most important and commonly used Git commands for easy reference.

INSTALLATION & GUIs

With platform specific installers for Git, GitHub also provides the ease of staying up-to-date with the latest releases of the command line tool while providing a graphical user interface for day-to-day interaction, review, and repository synchronization.

GitHub for Windows

<https://windows.github.com>

GitHub for Mac

<https://mac.github.com>

For Linux and Solaris platforms, the latest release is available on the official Git web site.

Git for All Platforms

<http://git-scm.com>

SETUP

Configuring user information used across all local repositories

```
git config --global user.name "[firstname lastname]"
```

set a name that is identifiable for credit when review version history

```
git config --global user.email "[valid-email]"
```

set an email address that will be associated with each history marker

```
git config --global color.ui auto
```

set automatic command line coloring for Git for easy reviewing

SETUP & INIT

Configuring user information, initializing and cloning repositories

```
git init
```

initialize an existing directory as a Git repository

```
git clone [url]
```

retrieve an entire repository from a hosted location via URL

STAGE & SNAPSHOT

Working with snapshots and the Git staging area

```
git status
```

show modified files in working directory, staged for your next commit

```
git add [file]
```

add a file as it looks now to your next commit (stage)

```
git reset [file]
```

unstage a file while retaining the changes in working directory

```
git diff
```

diff of what is changed but not staged

```
git diff --staged
```

diff of what is staged but not yet committed

```
git commit -m "[descriptive message]"
```

commit your staged content as a new commit snapshot

BRANCH & MERGE

Isolating work in branches, changing context, and integrating changes

```
git branch
```

list your branches. a * will appear next to the currently active branch

```
git branch [branch-name]
```

create a new branch at the current commit

```
git checkout
```

switch to another branch and check it out into your working directory

```
git merge [branch]
```

merge the specified branch's history into the current one

```
git log
```

show all commits in the current branch's history



INSPECT & COMPARE

Examining logs, diffs and object information

git log

show the commit history for the currently active branch

git log branchB..branchA

show the commits on branchA that are not on branchB

git log --follow [file]

show the commits that changed file, even across renames

git diff branchB..branchA

show the diff of what is in branchA that is not in branchB

git show [SHA]

show any object in Git in human-readable format

TRACKING PATH CHANGES

Versioning file removes and path changes

git rm [file]

delete the file from project and stage the removal for commit

git mv [existing-path] [new-path]

change an existing file path and stage the move

git log --stat -M

show all commit logs with indication of any paths that moved

IGNORING PATTERNS

Preventing unintentional staging or committing of files

```
logs/  
*.notes  
pattern*/
```

Save a file with desired patterns as .gitignore with either direct string matches or wildcard globs.

git config --global core.excludesfile [file]

system wide ignore pattern for all local repositories

SHARE & UPDATE

Retrieving updates from another repository and updating local repos

git remote add [alias] [url]

add a git URL as an alias

git fetch [alias]

fetch down all the branches from that Git remote

git merge [alias]/[branch]

merge a remote branch into your current branch to bring it up to date

git push [alias] [branch]

Transmit local branch commits to the remote repository branch

git pull

fetch and merge any commits from the tracking remote branch

REWRITE HISTORY

Rewriting branches, updating commits and clearing history

git rebase [branch]

apply any commits of current branch ahead of specified one

git reset --hard [commit]

clear staging area, rewrite working tree from specified commit

TEMPORARY COMMITS

Temporarily store modified, tracked files in order to change branches

git stash

Save modified and staged changes

git stash list

list stack-order of stashed file changes

git stash pop

write working from top of stash stack

git stash drop

discard the changes from top of stash stack

GitHub Education

Teach and learn better, together. GitHub is free for students and teachers. Discounts available for other educational uses.

✉ education@github.com
🌐 education.github.com

PROJECT MANAGEMENT ROADMAP

INTRODUCTION

Project management is a crucial skill in today's fast-paced world, whether you're planning a small event or overseeing a large project in a professional setting. This guide will provide you with a comprehensive roadmap to becoming an effective project manager, highlighting key concepts, methodologies, and tools.

WHAT IS PROJECT MANAGEMENT?

Project management involves using skills, tools, and techniques to create value through projects. It is a disciplined approach to planning, executing, and closing projects, ensuring that objectives are met within constraints such as time, budget, and resources.

KEY PHASES OF PROJECT MANAGEMENT

1. **Initiation:**
 - **Objective:** Define the project at a high level.
 - **Key Activities:** Secure authorization, identify stakeholders, create a project charter.
2. **Planning:**
 - **Objective:** Develop a detailed project plan.
 - **Key Activities:** Define scope, set objectives, create a schedule, identify resources, establish a budget, and develop subsidiary plans like risk management, communication, and quality management.
3. **Execution:**
 - **Objective:** Coordinate people and resources to implement the project plan.
 - **Key Activities:** Manage teams, communicate with stakeholders, ensure project activities are performed as planned.
4. **Monitoring and Controlling:**
 - **Objective:** Track, review, and regulate project progress and performance.
 - **Key Activities:** Measure performance against the project plan, identify variances, implement changes to keep the project on track.
5. **Closing:**
 - **Objective:** Finalize all project activities and deliver the final product.
 - **Key Activities:** Obtain client acceptance, close out contracts, document lessons learned, release project resources.

CHOOSING A PROJECT MANAGEMENT METHODOLOGY

Depending on the project type and industry, you might choose different project management methodologies:

- **Traditional (Waterfall) Project Management:**
 - **Characteristics:** Linear sequential approach, fixed scope, heavy documentation.
 - **When to Use:** Projects with well-defined requirements and clear stages.

- **Agile Project Management:**
 - **Characteristics:** Iterative, flexible approach, evolving requirements, continuous delivery of small functional increments.
 - **When to Use:** Projects requiring adaptability and frequent updates.

KEY METHODOLOGIES AND CONCEPTS

- **Scrum:**
 - **Role of Scrum Master:** Facilitates the Scrum process, removes impediments, ensures the team follows Agile practices.
 - **Key Elements:** Sprints, Daily Stand-ups, Backlog Grooming, Retrospectives.
- **Sprints:**
 - Time-boxed iterations, usually 1-4 weeks long, during which a specific set of work must be completed and made ready for review.

ESSENTIAL SKILLS AND TOOLS

- **Skills:**
- **Software Tools:**
 - **Microsoft Project:** Comprehensive project management software for planning, scheduling, and resource management.
 - **Jira:** Developed by Atlassian, used for issue and project tracking, particularly in software development.
 - **Smartsheet:** Work management and automation platform.

CERTIFICATIONS AND COURSES

- **Certifications:**
 - **PMP Certification:** Recognized globally, validates competence to perform in the role of a project manager. [Link](#)
- **Courses:**
 - **Google Project Management: Professional Certificate:** [Link](#)
 - **IBM Project Manager Professional Certificate:** [Link](#)
 - **Google Project Management Course:** [Link](#)

CAREER TIPS

- **Choosing the Right Industry:** Identify the industry (Tech, Finance, Healthcare, Manufacturing, Software Development) that aligns with your interests and skills.
- **Job Search Strategy:** Narrow down your job search to positions that match your exact skill set. It requires effort but increases the chances of finding a suitable role.

CONCLUSION

Project management is like being the super organizer of a big event, ensuring everything goes smoothly from start to finish. By understanding the key phases, choosing the right methodology, mastering essential skills, and leveraging appropriate tools, you can manage projects effectively and achieve your objectives.